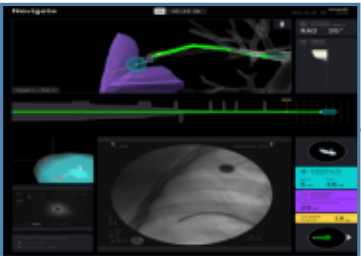




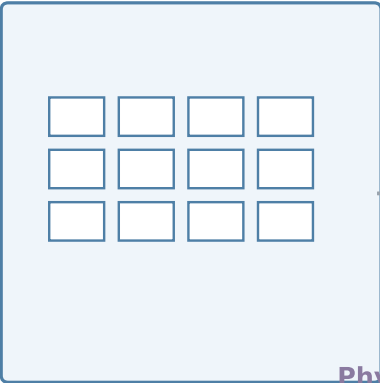
Laparoscopy



GI endoscopy



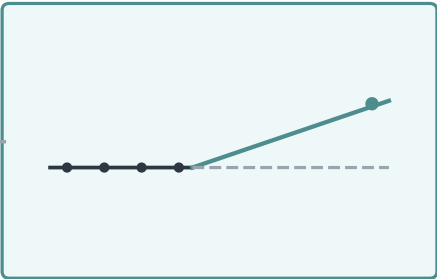
Bronchoscopy



Shared encoder
V-JEPA 2 ViT-L (frozen)

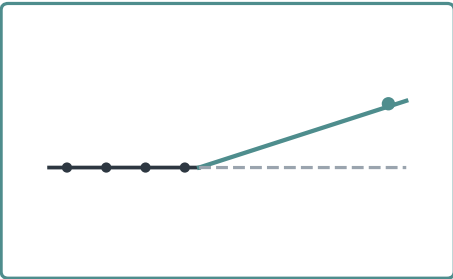
z

Passive-video forecast (validated)



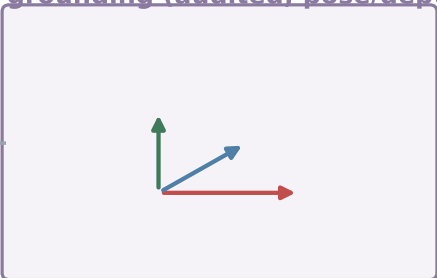
Causal residual L1 + coarse L2

forecast



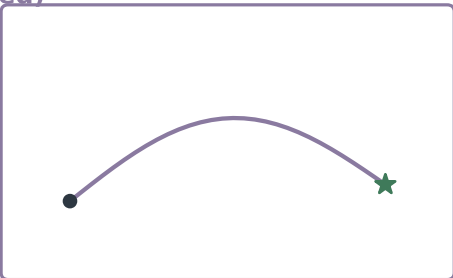
Capability 1: forecast evolution
0.978 cos, $p < 1e-80$

Physical grounding (audited: pose/depth-gated)



SE(3) block-causal dynamics

plan



Capabilities 2+3: 83.1% / 51.5%

Clinical application pathways

! Loss-of-view warning

◊ Camera-handling training

⤵ Navigation assistance

Input-sensitivity tests

history / action / domain

Reprojection pose gate

10.2 px -> 0.21 px

Matched negative banks

fixed, same-sequence

Grouped CV + frozen test

no test-set tuning